

A REVERSE COORDINATE INCLUSION OBSTRUCTION FOR A KALTON–PECK INCOMPARABILITY CONJECTURE

AGENT_LANE_03

ABSTRACT. Castillo and Moreno asked whether, for an interpolation pair (X^*, X) with a common unconditional basis, a continuous coordinate inclusion $X^* \rightarrow X$, $(X^*, X)_{1/2} = \ell_2$, and singular differential Ω , the spaces X and X^* must be incomparable. The wording is delicate because one coordinate inclusion is already assumed. This note proves a precise partial result: under the standard complex/CKMR-derived setting used in the source paper, singularity rules out the reverse coordinate inclusion $X \rightarrow X^*$. Equivalently, the two endpoint norms cannot be mutually comparable on the common coordinate space. This does not rule out arbitrary non-coordinate Banach-space embeddings.

1. SOURCE QUESTION

The source paper is Castillo–Moreno, *The Kalton–Peck space is the complexification of the real Kalton–Peck space*, arXiv:2204.00672, published in J. Math. Anal. Appl. 515 (2022), Article 126374.

The relevant passage is in Section 3, “Concluding remarks”, on page 9 of the arXiv PDF:

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3. Concluding remarks

Having singular differential is a rather demanding condition. For instance:

PROPOSITION 3. *Let (X, X^*) be a real or complex interpolation pair of Banach spaces with a common unconditional basis for which there exists a continuous inclusion $X^* \rightarrow X$ and such that $(X^*, X)_{1/2} = \ell_2$ with differential Ω . If Ω is singular then X^* does not contain ℓ_2 .*

PROOF. Indeed, let (u_n) be blocks in X^* so that $[(u_n)] \simeq \ell_2$. Pick $u \in [(u_n)]$ and observe that

$$\begin{aligned} \|u\|_X &= \sup\{\langle y, u \rangle : \|y\| \leq 1; y \in X^*\} \\ &\geq \sup\{\langle y, u \rangle : \|y\| \leq 1 : y \in [(u_n)]\} \\ &= \|u\|_{[(u_n)]^*} \sim \|u\|_2. \end{aligned}$$

Since $\|u\|_X \leq \|u\|_2$ it turns out that $\|u\|_{X^*} \sim \|u\|_2$. Thus, the norms of X and X^* are equivalent on $[(u_n)]$, and this obliges $\Omega|_{[(u_n)]}$ to be bounded. $\square$

Optimistic readers could now easily believe the following conjecture:

Conjecture. *Let (X, X^*) be an interpolation pair of Banach spaces with a common unconditional basis for which there exists a continuous inclusion $X^* \rightarrow X$ and such that $(X^*, X)_{1/2} = \ell_2$ with differential Ω . If Ω is singular then X and X^* are incomparable.*

We do not have a proof for that. Optimistic readers should be warned: an example in [5] provides two incomparable spaces (A, B) not containing ℓ_2 so that the complex differential Ω at $1/2$ is an isomorphism on a complemented copy of ℓ_2 . Thus, the converse for the assertion above and for that in Proposition [3] fail.

In text form, the conjecture says:

Let (X, X^*) be an interpolation pair of Banach spaces with a common unconditional basis for which there exists a continuous inclusion $X^* \rightarrow X$ and such that $(X^*, X)_{1/2} = \ell_2$ with differential Ω . If Ω is singular then X and X^* are incomparable.

Here singularity means that the associated exact sequence has strictly singular quotient map; equivalently, the restriction of Ω to any infinite-dimensional subspace is never a sum of a bounded map and a linear map.

2. PROOF INTUITION

The conjecture asks for incomparability, but the hypotheses already include one coordinate inclusion, $X^* \rightarrow X$. Thus the immediate consistent coordinate reading is: can the reverse coordinate inclusion $X \rightarrow X^*$ also hold?

It cannot. If both coordinate inclusions are bounded, the two endpoint norms are equivalent. An interpolation couple with equivalent endpoints is, up to equivalent couple norms, just the diagonal couple (E, E) . For a diagonal couple one can select an interpolation representative by multiplying the vector by a fixed scalar extremal; the derivative of that selector is a fixed scalar multiple of the vector. Hence the differential is linear, modulo the usual bounded selector-change error. A globally linear-plus-bounded differential is not singular.

3. MAIN RESULT

Theorem 1. *Let (X^*, X) be a compatible interpolation couple with a common unconditional coordinate basis and suppose the interpolation method is one of the usual complex methods, or a CKMR*

pseudolattice method admitting differentiation, as in the source paper. Assume

$$(X^*, X)_{1/2} = \ell_2$$

with associated differential Ω , and assume that the coordinate inclusion $X^ \hookrightarrow X$ is bounded. If Ω is singular, then the reverse coordinate inclusion*

$$X \hookrightarrow X^*$$

is not bounded.

Proof. We prove the contrapositive. Suppose that the reverse coordinate inclusion is also bounded. Then there are constants $a, b > 0$ such that for all finitely supported coordinate vectors u ,

$$\|u\|_X \leq a\|u\|_{X^*} \quad \text{and} \quad \|u\|_{X^*} \leq b\|u\|_X.$$

Thus the two endpoint norms are equivalent on the common coordinate core and the two completed endpoint spaces can be identified as the same Banach space E , with equivalent norms.

Interpolation is stable under equivalence of compatible couples. Therefore $(X^*, X)_{1/2}$ is, through the coordinate identity, equivalent to the diagonal interpolation space $(E, E)_{1/2}$. The diagonal interpolation space is just E , with equivalent norm. Since the hypothesis identifies $(X^*, X)_{1/2}$ with ℓ_2 , it follows that the endpoint norm of E is equivalent to the ℓ_2 norm on the interpolation domain.

It remains to inspect the derived map. For the diagonal couple (E, E) choose one fixed scalar admissible representative $\lambda = (\lambda_n)$ for the scalar unit at the parameter $1/2$; in the complex method this is simply the constant analytic function 1, while in the CKMR description one may take any finite scalar sequence satisfying

$$\sum_n e^{n/2} \lambda_n = 1.$$

For $x \in E$ put $Sx = (\lambda_n x)_n$. Because the same norm is used at both endpoints, $S : E \rightarrow \mathcal{J}((E, E))$ is bounded. Its derivative is

$$\delta'_{1/2} Sx = \left(\sum_n n e^{(n-1)/2} \lambda_n \right) x,$$

or 0 in the constant-function complex model. In either description this is a bounded linear map on $E \simeq \ell_2$.

Transporting this selector through the endpoint norm equivalence gives a bounded selector for the original couple (X^*, X) whose differential is bounded linear on ℓ_2 . The usual selector-independence statement for derived interpolation spaces says that replacing one bounded selector by another changes the differential only by a bounded map into the interpolation space. Hence the source differential Ω is a sum of a bounded map and a linear map on all of ℓ_2 .

This is incompatible with singularity: a singular differential has no infinite-dimensional restriction which is linear plus bounded. Therefore the reverse coordinate inclusion cannot have been bounded. $\square$

Corollary 1. *Under the hypotheses above, singularity prevents mutual coordinate comparability of X and X^* : the assumed inclusion $X^* \hookrightarrow X$ cannot be accompanied by a bounded coordinate inclusion $X \hookrightarrow X^*$.*

4. LIMITATIONS AND NOVELTY CHECK

This is a partial result, not a full solution of the broadest possible reading of the conjecture. It rules out the reverse coordinate inclusion on the common basis. It does not rule out non-coordinate isomorphic embeddings of X into X^* , embeddings of X^* into X with nonclosed range, or stronger notions of Banach-space incomparability.

Bounded novelty checks were performed before promotion. The lightweight run indexes had no exact entry for arXiv:2204.00672. Web searches for the phrases “2204.00672 singular differential

incomparable”, “The Kalton–Peck space is the complexification incomparable”, and “If Omega is singular then X and X^* are incomparable” found the source arXiv/JMAA paper and nearby Kalton–Peck literature, but no later paper explicitly resolving the conjecture. The 2024 paper *Interpolator Symmetries and New Kalton–Peck Spaces* discusses related Kalton–Peck interpolation spaces and open questions, but not this reverse-coordinate obstruction.

5. HUMAN REVIEW RECOMMENDATION

Review as a likely valid partial theorem. The key point to check is the selector-independence step, and the functoriality step, for the exact real or complex interpolation framework intended by the source conjecture. The claim should not be upgraded to a full solution unless “incomparable” is explicitly interpreted as failure of mutual coordinate inclusions.

REFERENCES

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- [3] N. J. Kalton and N. T. Peck, *Twisted sums of sequence spaces and the three-space problem*, Trans. Amer. Math. Soc. 255 (1979), 1–30.
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